

JORDAN B. S. VAN BEECK

Asteroseismology | Stellar Astrophysics | FOSS



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SCIENTIFIC EDUCATION

Ph.D., Astronomy & Astrophysics, KU Leuven (Belgium), September 2023

Asteroseismology of Kepler B stars: internal magnetism and non-linear mode coupling

Promoters: Prof. Dr. C. Aerts, Prof. Dr. T. Van Hoolst, Prof. Dr. D. M. Bowman

M.Sc., Astronomy & Astrophysics, KU Leuven (Belgium), July 2019 – *Magna Cum Laude*

The influence of an interior magnetic field on gravity-mode oscillations of intermediate-mass stars

Promoters: Prof. Dr. C. Aerts, Dr. T. Van Reeth, Prof. Dr. D. M. Bowman

M.Sc., Chemistry, University of Antwerp (Belgium), September 2017 – *Magna Cum Laude*

Characterization of radioactive particles

Promoters: Prof. Dr. K. Janssens, Prof. Dr. B. Salbu, Prof. Dr. O.-C. Lind, M.Sc. G. Nuyts

B.Sc., Chemistry, University of Antwerp (Belgium), September 2015 – *Magna Cum Laude*

Atomic scale reactive MD studies of DNA oxidation for plasma oncology: the role of H₂O₂ and HO₂

Promoters: Prof. Dr. A. Bogaerts, Prof. Dr. E. Neyts, Dr. C. Verlackt

EMPLOYMENT

Heidelberg Institute for Theoretical Studies (HITS), TOS group, Heidelberg, Germany

Postdoctoral Researcher Asteroseismology February 2025 – March 2027

Project: higher-order rotational effects on mixed-mode oscillations of red giants

KU Leuven, Department of Physics and Astronomy, Institute of Astronomy, Leuven, Belgium

Research Associate, Asteroseismology, September 2023 – December 2023

Project: non-linear asteroseismic modeling of slowly pulsating B stars

AWARDS AND FELLOWSHIPS

2021 - 2022	FWO(Research Foundation Flanders) long research stay grant (€12 738) for a stay as a VSR at TAPIR, Caltech, CA, USA with Prof. Dr. J. Fuller.
2019 - 2023	4-year PhD Fellowship, Department of Physics and Astronomy, KU Leuven
2019	Paul Smeyers Prize, KU Leuven Awarded to the annual best master's thesis in Astronomy & Astrophysics at the June examination session.

TEACHING EXPERIENCE

Lecturer, Heidelberg University

2025	Theoretical introduction for 'Asteroseismology' (Master-level physics course that introduces various aspects of asteroseismology.)
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Teaching Assistant, Department of Physics and Astronomy, KU Leuven

2022 - 2023	K01B4A: 'Natuurkunde met elementen van wiskunde' I (taught in Dutch) (Introductory physics course for the first-year students in the B.Sc. of pharmaceutical sciences)
2019 - 2021	K01B4A B: 'Natuurkunde met elementen van wiskunde' I II (taught in Dutch) (Introductory physics courses for the first-year students in the B.Sc. of pharmaceutical sciences) English translation of course titles: 'Physics with elements of mathematics' I II

VISITING RESEARCH POSITIONS (ABRIDGED)

11/2021 - 06/2022	Visiting Student Researcher (VSR) at TAPIR, Caltech, CA, USA Project goal: development of computational tools that aid in explaining oscillation mode amplitude limitation in slowly pulsating B stars Mentor: Prof. Dr. J. Fuller <i>Funding: FWO long research stay grant</i>
11/2021 - 12/2021	Transport in Stellar Interiors Program Affiliate (workshop, conference) at KITP, UC Santa Barbara, CA, USA Mentor: Prof. Dr. L. Bildsten <i>Funding: FWO long research stay grant</i>

SEMINARS/CONFERENCE PRESENTATIONS (ABRIDGED)

July 2026	POSTER PRESENTATION (doi:10.5281/zenodo.21083361) TASC10 / KASC17 workshop/conference at Aarhus University, Denmark.
July 2025	CONTRIBUTED TALK (doi:10.5281/zenodo.17241657) TASC9 / KASC16 workshop/conference at ISTA, Austria.
March 2023	INVITED SEMINAR (https://youtu.be/dWaEkri_s2I?feature=shared) The Good Vibrations seminar series, S3E8.
February 2023	INVITED SEMINAR (https://youtu.be/nTWZoLCMa_g?feature=shared) Ad-valvas seminar of the Institute of Astronomy, KU Leuven, Belgium.
July 2022	POSTER PRESENTATION (additional info: doi:10.5281/zenodo.6814504) TASC6 / KASC13 workshop/conference at KU Leuven, Belgium.
November 2021	CONTRIBUTED TALK (doi:10.26081/K6VH15) Transfer in Stellar Interiors conference (part of the Probes of Transport in Stars program), Kavli Institute for Theoretical Physics (KITP), UC Santa Barbara, CA, USA.

PROFESSIONAL ACTIVITIES

Conference organization

July 2022	Member of the Local Organizing Committee TASC6 / KASC13 workshop/conference in Leuven, Belgium
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Professionally associated as

2026 – ...	Junior Member of the <i>International Astronomical Union (IAU)</i>
2022	Fellow (Graduate Student) of the <i>Royal Astronomical Society (RAS)</i>
2020 – 2022	Graduate Student Member of the <i>American Astronomical Society (AAS)</i>
2020 – 2022	Graduate Student Member of the <i>Royal Netherlands Astronomical Society (KNA)</i>
2019 – 2023	Graduate Student Member of the <i>International Research Network for Nuclear Astrophysics (IReNA)</i>

DIGITAL PROFICIENCY

